

Students Records

These describe student records with their ID, full name (last name first), and grades for 4 assessments (assignment 1, assignment 2, assignment 3, and test):

1. Student ID **1001: Alice Allteron**
Grades: Assignment 1 = 92, Assignment 2 = 75, Assignment 3 = 100, Test = 85
2. Student ID **1002: Bob Boberson**
Grades: Assignment 1 = 50, Assignment 2 = 64, Assignment 3 = 73, Test = 66
3. Student ID **1003: Charlie Charleston**
Grades: Assignment 1 = 65, Assignment 2 = 75, Assignment 3 = 85, Test = 100
4. Student ID **1004: Dave Davidson**
Grades: Assignment 1 = 10, Assignment 2 = 92, Assignment 3 = 34, Test = 87
5. Student ID **1005: Ellen Ellenovitch**
Grades: Assignment 1 = 72, Assignment 2 = 87, Assignment 3 = 82, Test = 70



Prolog knowledge base

```
% student(Id, name([Last, First]), grades([A1, A2, A3, Test])).  
student(1001, name([allteron, alice]), grades([92, 75, 100, 85])).  
student(1002, name([boberson, bob]), grades([50, 64, 73, 66])).  
student(1003, name([charleston, charlie]), grades([65, 75, 85, 100])).  
student(1004, name([davidson, dave]), grades([10, 92, 34, 87])).  
student(1005, name([ellenovitch, ellen]), grades([72, 87, 82, 70])).
```

prolog

```
% student(Id, name([Last, First]), grades([A1, A2, A3, Test])).  
  
student(1001, name([allteron, alice]), grades([92, 75, 100, 85])).  
student(1002, name([boberson, bob]), grades([50, 64, 73, 66])).  
student(1003, name([charleston, charlie]), grades([65, 75, 85, 100])).  
student(1004, name([davidson, dave]), grades([10, 92, 34, 87])).  
student(1005, name([ellenovitch, ellen]), grades([72, 87, 82, 70])).
```

Queries

1. How did Alice do on assignment 1?

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?- student(_, name([allteron, alice]), grades([A1 | _])).
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2. How did Alice do on assignment 2?

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?- student(_, name([allteron, alice]), grades([_, A2 | _])).
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3. How did student 1002 do on the test?

?- student(1002, _, grades([_, _, _, Test])).

4. Who got 100 on the test?

?- student(_, name(Name), grades(L), last(L, Test), Test == 100.

5. Did anyone get 100 on anything (any assignment or test)?

?- student(_, name(Name), grades(L), member(100, L).

6. What is each student's average grade (equal weighting)?

?- student(_, name(Name), grades([A1, A2, A3, Test])), Average is (A1 + A2 + A3 + Test) / 4.